

Banula Kumarage

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Summary

An enthusiastic individual seeking a position where I meet new challenges every day, utilizing my fullest potential while getting a chance to work with new technologies. I am a hardworking person with the capability and skills to adapt to a new working environment, aiming to contribute my fullest effort to achieve goals.

Working Experience

Software Engineer, Sysco LABS Sri Lanka June 2024 – Present

- Working on Backend Development, Cloud Development, and DevOps tasks
- Developing Kafka stream processing applications with Spring Boot and Apache Beam.
- **Technologies:** Java (Spring Boot Framework), AWS, Kafka, Apache Beam, PostgreSQL, Redis, Elasticsearch

Software Engineer Intern, IFS R&D International (Pvt.) Ltd Jan 2023 – June 2023

- Worked on feature enhancements and fixing defects. Worked on CI/CD tasks which automated the manual tasks. Worked on Helm Chart Improvements of the micro-services. Worked on performance testing tasks.
- **Technologies:** Java (Quarkus Framework), PL/SQL, Azure, Kubernetes, Docker, Jenkins, Helm

Education

BSc. Eng. (Hons) Specialized in Computer Science & Engineering 2020 - 2024

Department of Computer Science and Engineering, University of Moratuwa

- Overall GPA: First Class GPA of 3.96 out of 4.2
- Dean's List in semesters 1, 2, 3, 4, 5, 6, 7, 8

G.C.E Advanced Level Examination - Physical Science Stream 2018

Central College, Anuradhapura

- Island Rank - 173rd
- District Rank - 4th
- Z-Score - 2.2588

Publications

Use of Federated Learning for Personal Smart Media Cloud Solutions Aug 2004

Banula Kumarage, Hiruna Harankahadeniya, Akila Induranga, Indika Perera, Kutila Gunasekera

10.1109/MERCon63886.2024.10688883

Projects

PicsSmart : Privacy Ensured Smart Media Cloud with Federated Learning July 2023 - June 2024

- An edge-processing-based application where users are concerned about their photos' privacy can still benefit from ML-powered smart features such as face-identification, face-clustering, and image caption generation. For the models to work well with heterogeneous datasets, models are to be frequently trained at the edge and updated using Federated Transfer Learning techniques.
- **Contribution:** Developing backend server with smart features, Federated Learning
- **Technologies:** FastAPI, ReactJS, LAVIS, Keras, Tensorflow Federated, Docker

Scatterrr : Secure and trusted distributed file storage system April 2024 - July 2024

- A secure distributed file storage system that segments files into equal-sized chunks and distributes them randomly across a cluster of nodes for enhanced security. Using a Merkle Tree, it ensures data integrity, allowing users to verify files remain tamper-free. The system offers an intuitive interface for uploading, searching, viewing

metadata, verifying integrity, and downloading files, combining reliability with robust security principles.

- **Contribution:** Back-end Developing
- **Technologies:** Java(Spring Boot), TypeScript(Next.js)

Streamix : Dash compliant video streaming system

Nov 2022 - Dec 2022

- The project is a dash (Dynamic Adaptive Streaming over HTTP) video streaming system. Android app is used to capture the video and upload it to the centralized server. Anyone connected to the server can watch the uploaded videos with different qualities.
- **Contribution:** Back-end Developing
- **Technologies:** Java, Node.JS, MongoDB, FFmpeg

Moodle plugin for establishing workflows

Sep 2022 - Nov 2022

- The purpose of the project is to create a plugin to automate the process of submitting and validating student requests regarding various Moodle activities.
- **Contribution:** As both a Back-end and Front-end developer
- **Technologies:** PHP, MySQL, JS, Mustache

Supply Chain Management System

June 2022 - July 2022

- The project is an information and data management system based on the distribution flow of a company. The complete chain of the distribution flow from the store to the customer can be monitored using the system.
- **Contribution:** As both a Back-end and Front-end developer
- **Technologies:** PHP, MySQL, JS, HTML

Pharmacy Management System - Home Pharmacist

Dec 2021 - Jan 2022

- A centralized system for pharmacies to provide many facilities including searching and ordering from pharmacies that are available in the system.
- **Contribution:** As both a Back-end and Front-end developer
- **Technologies:** HTML, CSS, PHP, MySQL

Student Request Management System

Dec 2021 - Jan 2022

- A web-based management system to manage student requests which have been sent to the academic staff in the university.
- **Contribution:** Back-end Developing
- **Technologies:** Node.JS, Express, React, MongoDB

Skills

Technical Skills

- Java | Python | C++ | Javascript | Typescript | PHP
- Spring Boot | Django | Quarkus | FastAPI | Node.js | Express | React JS
- MySQL | MongoDB | PostgreSQL | PL/SQL | SQLite
- Kubernetes | Helm | Docker | Kafka | Apache Beam | Redis | Elasticsearch
- AWS | Azure | GCP

Non Technical Skills

- Leadership | Teamwork | Communication | Public Speaking | Analytical Thinking

Certifications

Introduction to GCP Infrastructure

2024

- Obtained a verified badge after completing the lab tasks offered by Google Cloud

Application Development using Microservices and Serverless

2024

- Obtained a verified badge after completing the course offered by Coursera

AZ-900: Microsoft Azure Fundamentals Certification

2023

- Obtained a verified certificate after completing the certification exam offered by Microsoft

Kubernetes Certified Application Developer (CKAD) with Tests	2023
Completed a course on the Udemy platform offered by KodeCloud	
Neural Networks and Deep Learning	2023
Completed a course on the Coursera platform offered by DeepLearning.AI	
Introduction to Data Science in Python	2021
Completed a course on the Coursera platform offered by the University of Michigan.	

Achievements

Virtual Colleagues' Choice Award for the FY25 Sysco Technology Hackathon	Oct 2024
Our team ByteChef has won the Virtual Colleagues Choice Award in the Sysco Global Hackathon 2024! Also, we've been selected as a Top 5 finalist for our Generative AI-based solution for boosting the productivity of our sales reps.	
Winners - RealHack 5.0	April 2024
In the 24-hour hackathon organized by the University of Kelaniya, our implemented system of Multi-modal authenticity validation system for virtual interviews secured first place.	
IEEEExtreme 16.0 - 24-hour Algorithmic Coding Competition	Oct 2022
Team404 World Rank – 300 Country Rank – 12	

Responsibilities

Teaching Assistant
Computer Networks (CS3033), Data Communication (CS2033) - Department of Computer Science & Engineering, University of Moratuwa
Deputy Head - Industrial Relations Committee and Department Coordinator
EXMO 2023 Technical Exhibition organized by the University of Moratuwa
Field Representative
Department of Computer Science and Engineering - batch 19 - Semester 3&4

Organizations

Toastmasters International	03/2019 - Present
Member no: 7317863	
Institute of Electrical and Electronic Engineers (IEEE)	2021 - Present

References

- Available on Request